University Of Michigan – Dearborn
Isolated Bidirectional DC-DC Converter Design
ECE517 - Term Project – Fall 2024

STUDENT NAME: HAMZA AL-ZAKARNEH

DATE: 12/09/2024

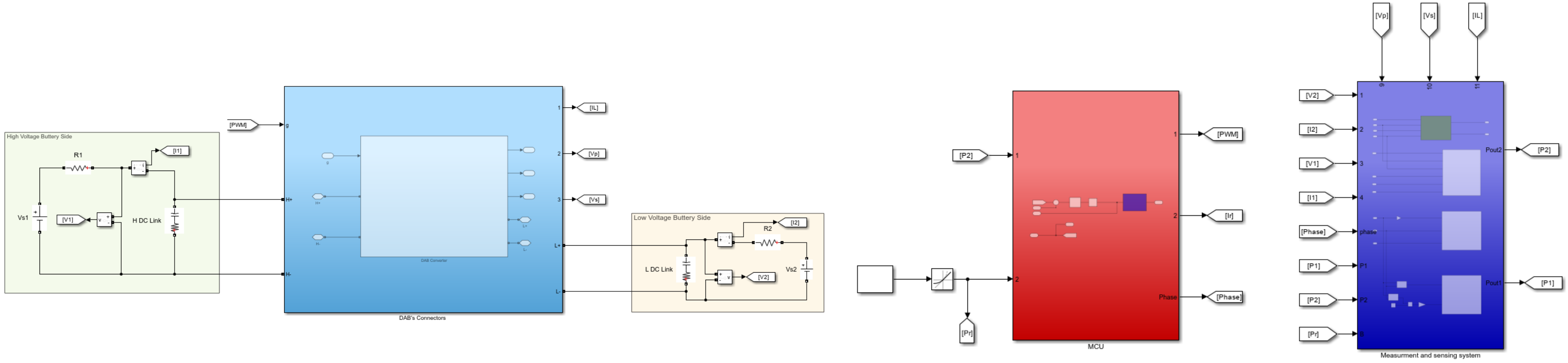
Introduction

- The project objective is to design a high efficiency isolated bidirectional DC-DC converter for distributed energy storage devices (batteries). The Dual Active Bridge (DAB) converter was used in this project to distribute the energy between 400 V and 45 V batteries, with efficiency higher than 95% at 2kW full load.
- The Dual Active Bridge (DAB) converter is one of the common isolated bidirectional DC-DC converters topologies. It is known for its high efficiency and power density
- Dual Active Bridge converters with the traditional PID controller are designed in this project to distribute power up to 2kW between 400 V and 45 V batteries, with output ripples less than 3% and efficiency higher than 96%.

Model Design

Model Design

- Top Level Schematics:



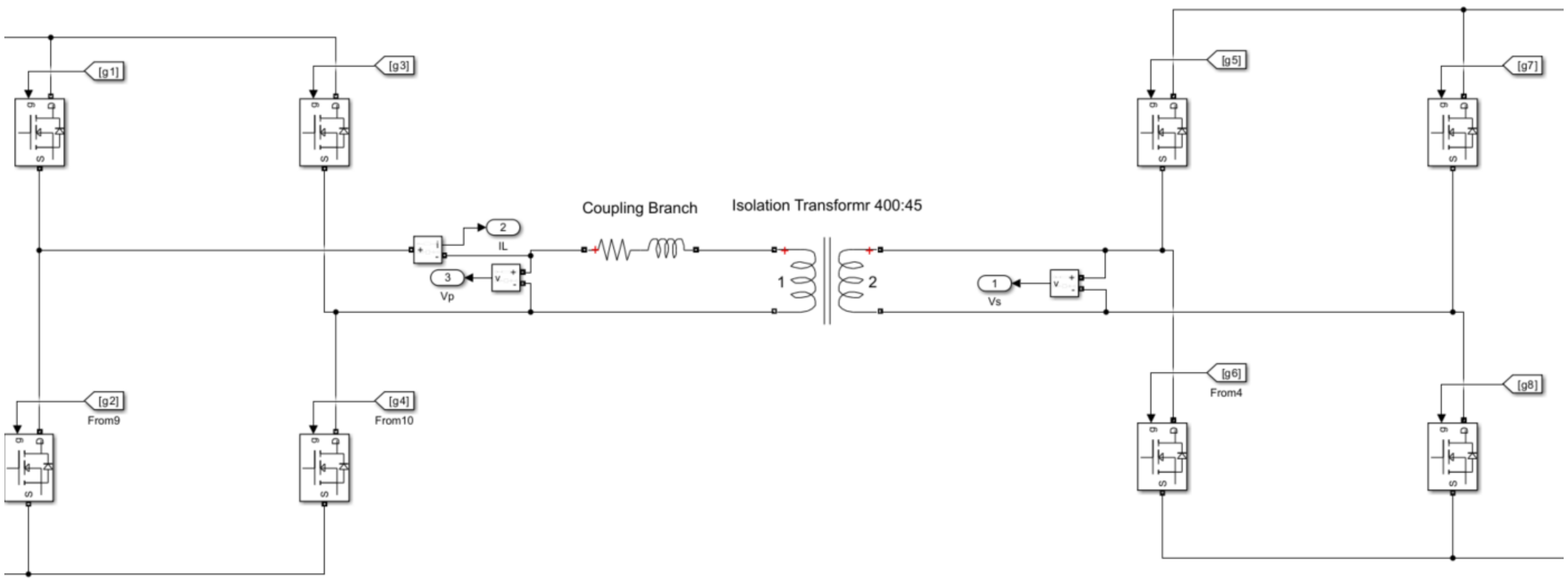
Model Design

- Voltage Battery Sides:

1. 400V, 2kW battery - HDL3000PS400 400V battery with 20mΩ internal resistance can be used.
2. 45V, 2kW battery - RSP-2000-48 45V battery with 60mΩ internal resistance can be used.
3. DC links to filter the noise, store the energy and stabilize the voltage in each side.
   1. The required capacitor dc link can be calculated using *Equation 1.*
   2. $C \geq \frac{Iripple}{4\ f\ \Delta V} \rightarrow C \geq \frac{1.6}{4\ 50\ 12} = 667\mu F$ *(Equation 1)*
   3. KEMET ALS70A102NF450 1mF with 1mΩ ESR resistance was used in the design.

Model Design

- DAB converter:



Model Design

- DAB converter:

1. Typical type of transistors can be used to design the two full bridges in the DAB converter, the transistor must handle voltage higher than 400V. IPP110N20N3/IPW60R037P7 transistors with $11m\Omega$ transistor on-state resistance, $1M\Omega$ Snubber resistance Rs, $10m\Omega$ and 0.9 diode voltage drop were utilized to build the two full bridges.

Model Design

- DAB converter:

2. The suitable DAB coupling inductance can be calculated using *Equation 2*, which represents the inductance that is required to generate the maximum power, such an inductance can be found in different supplier websites. 744355110 is an example of 100µH inductance with a 4.5mΩ DCR resistance.

$$L \geq \frac{V_1 \, V_2}{2 \, \pi \, f \, P_{max}} \rightarrow L \geq \frac{400 * 45}{2 \, \pi \, 50k \, 2k} = 30\mu H. \quad (Equation\ 2)$$

Model Design

- DAB converter:

3. *Table 1* shows the parameters that were chosen to design the 400/45 transformer. This type of transformer can be designed by Infineon Technologies

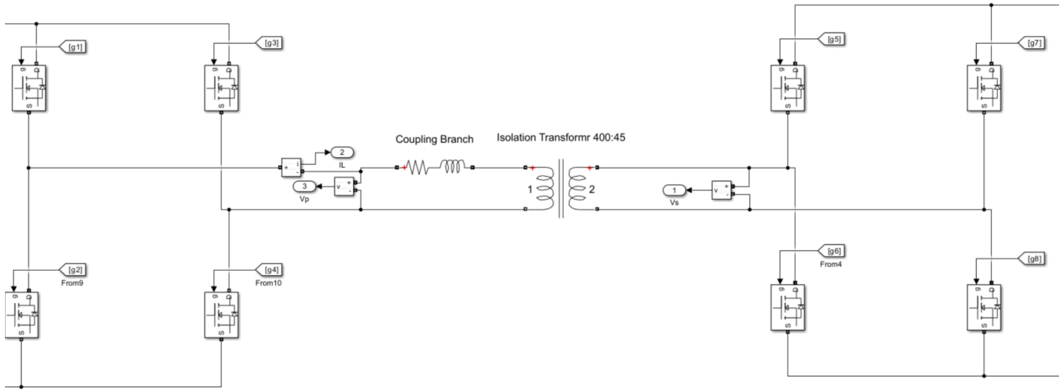
Parameter	Value	Comments
Rated Power	2kW	This value was given by the project's requirements. However, higher power rate is preferred.
Switching Frequency	50 k (Hz)	Frequencies around 50–100 kHz are common, balancing efficiency and transformer size.
Primary side resistance	10mΩ	The system works fine even if the resistance is increased up to 50m Ohm.
leakage inductance	15µH	Transformer with leakage inductance between 10-15 µH is common
Magnetization resistance	10mΩ	Resistance can be increased to reduce the magnetization current. However, after a certain value the loss can be increased because of high resistance value.
Magnetization inductance	16mH	Common in transformers

Model Design

- Control System Design:

By controlling the full bridge gates, a phase shift between the primary side and the secondary side voltage will be added and the power will be transferred from the leading side to the lagging one according to *Equation 3*.

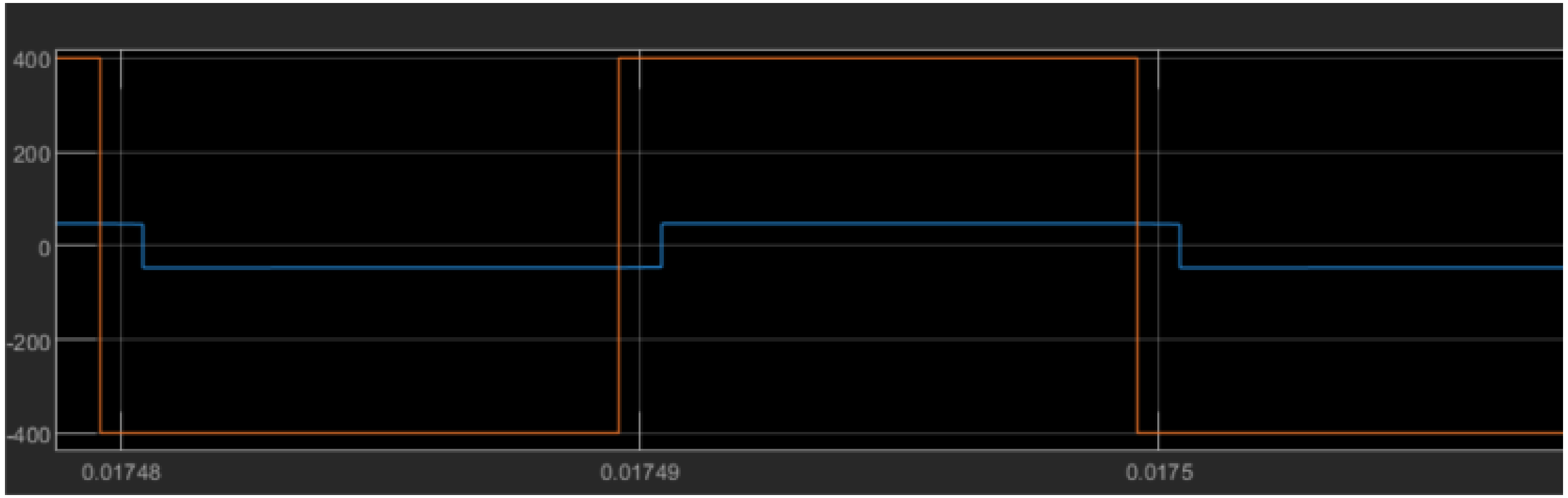
$$Power = \frac{V_1 V_2}{X_L} \, f(phase\ shift) \qquad (Equation\ 3)$$



Model Design

- Control System Design:

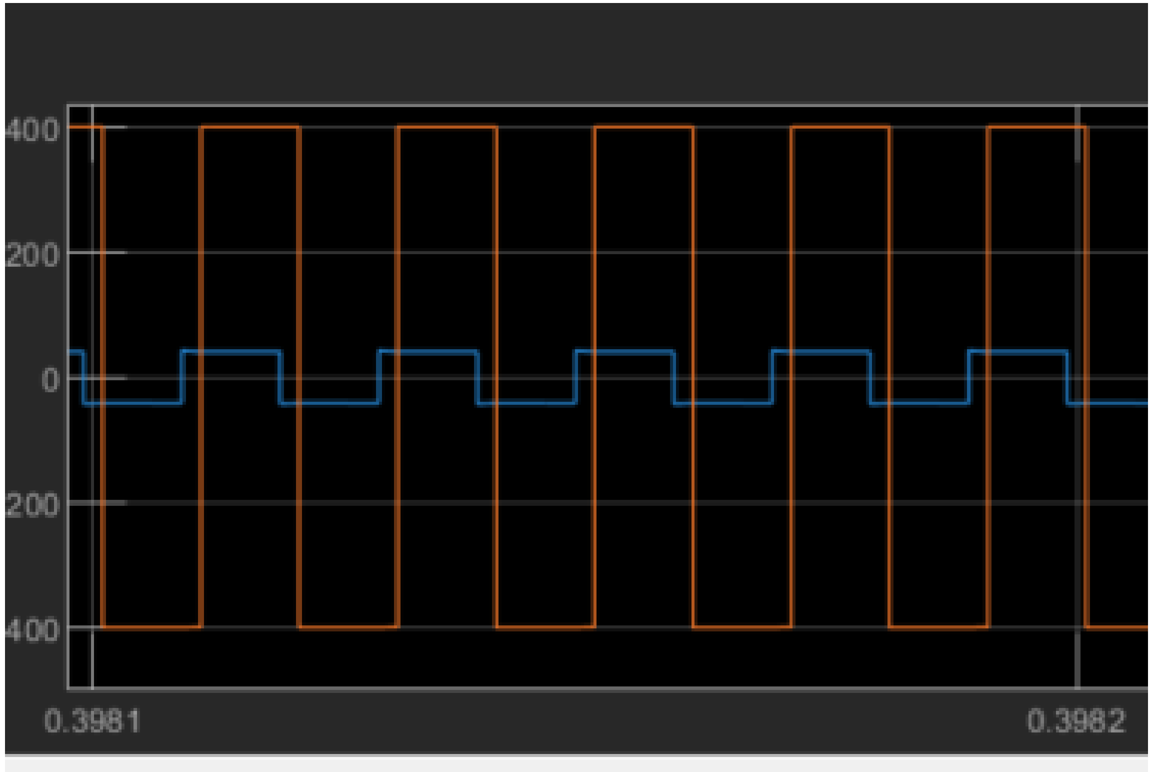
Transformer primary and secondary voltage during discharging energy from the high voltage side to the low voltage side



Model Design

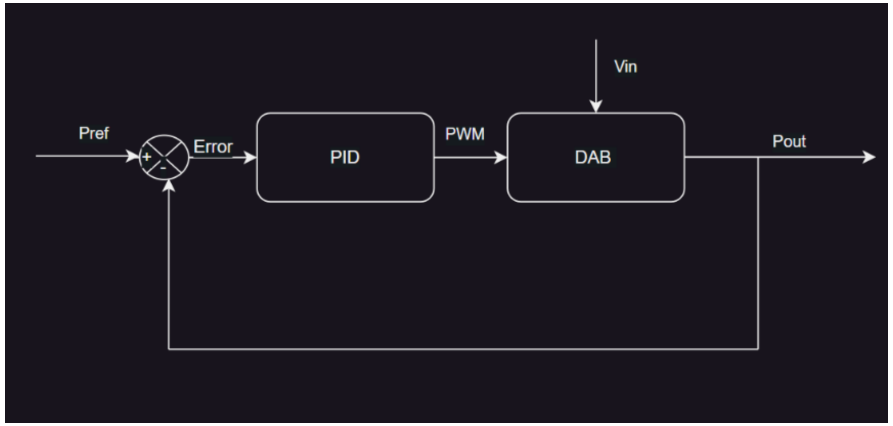
- Control System Design:

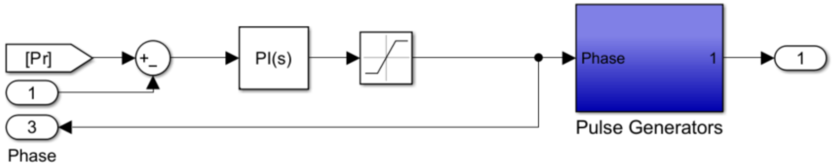
Transformer primary and secondary voltage during discharging energy from the low voltage side to the high voltage side



Model Design

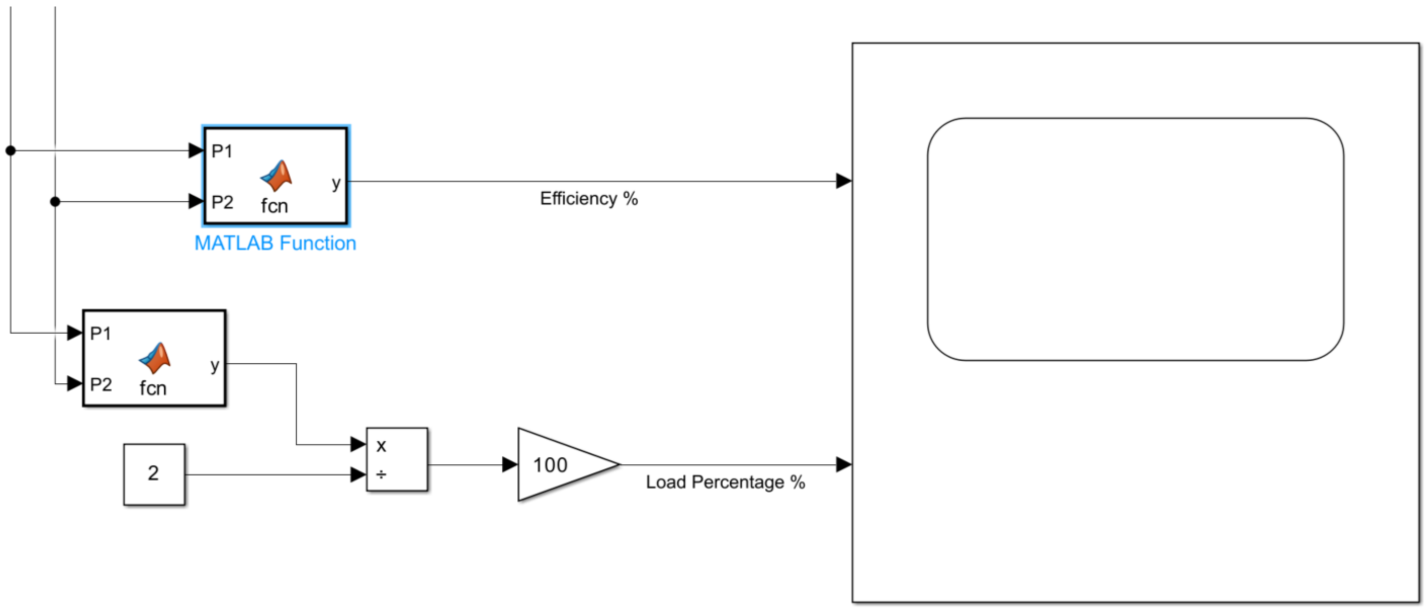
- Control System Design:

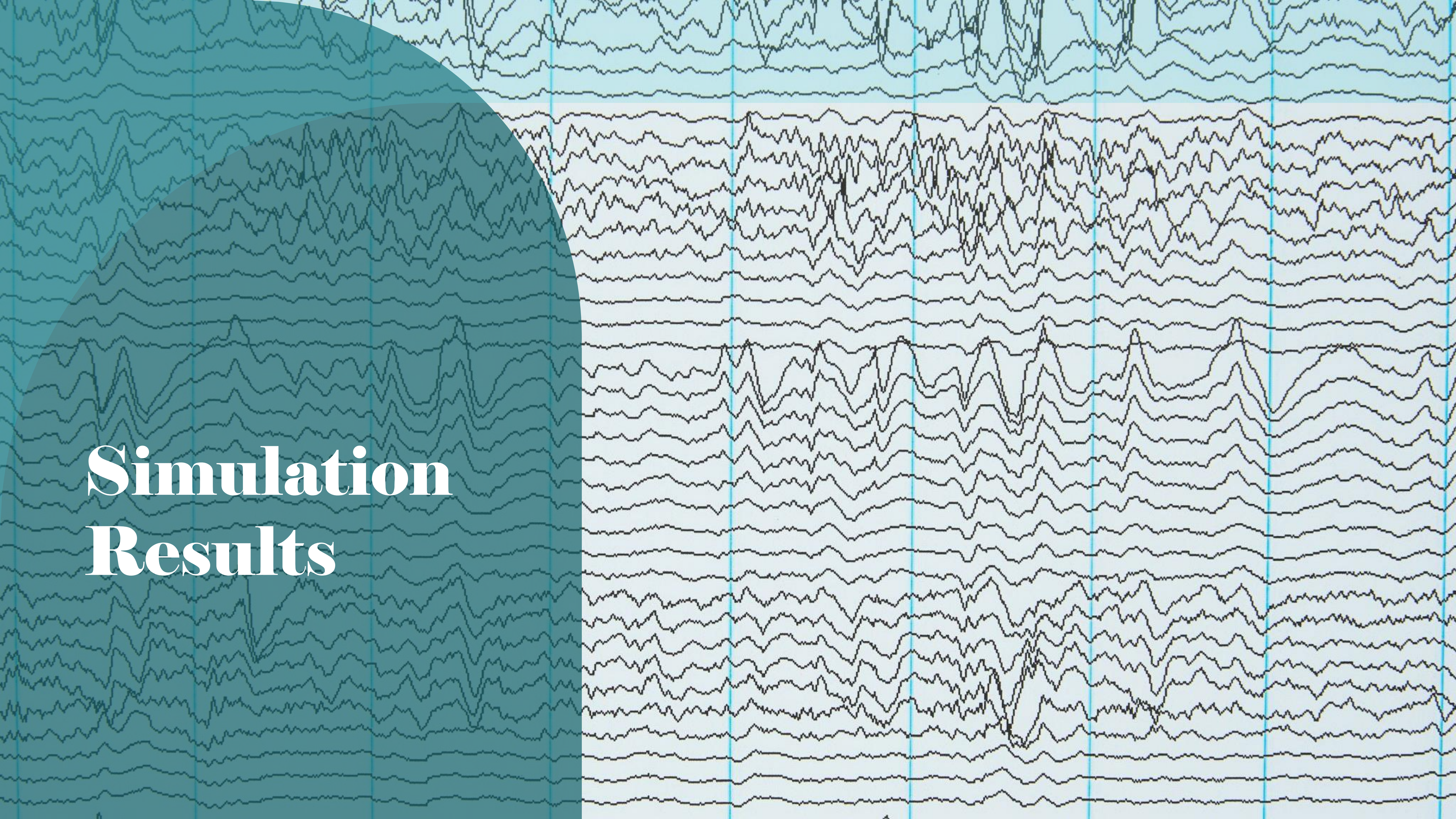




Model Design

- **Measurements and Sensing system:** consists of Power measurement subsystem and Scopes to plot the outputs

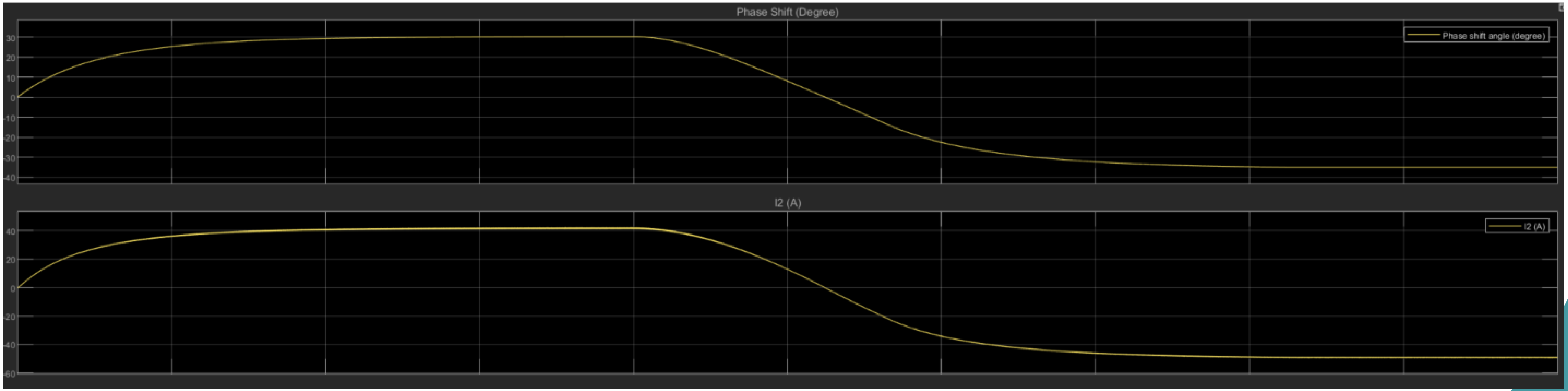




Simulation Results

Simulation Results

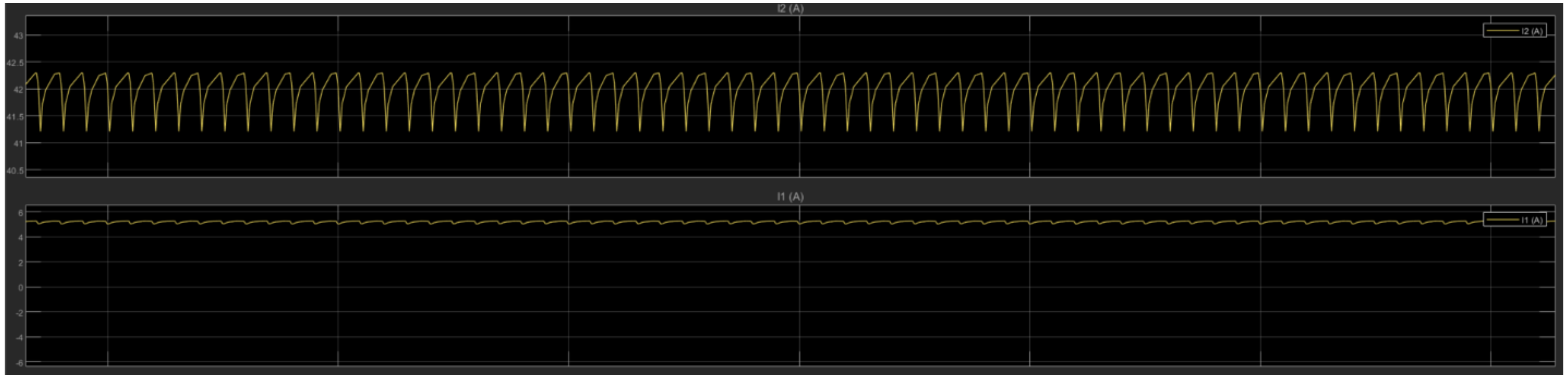
Phase angle and current curves



Simulation Results

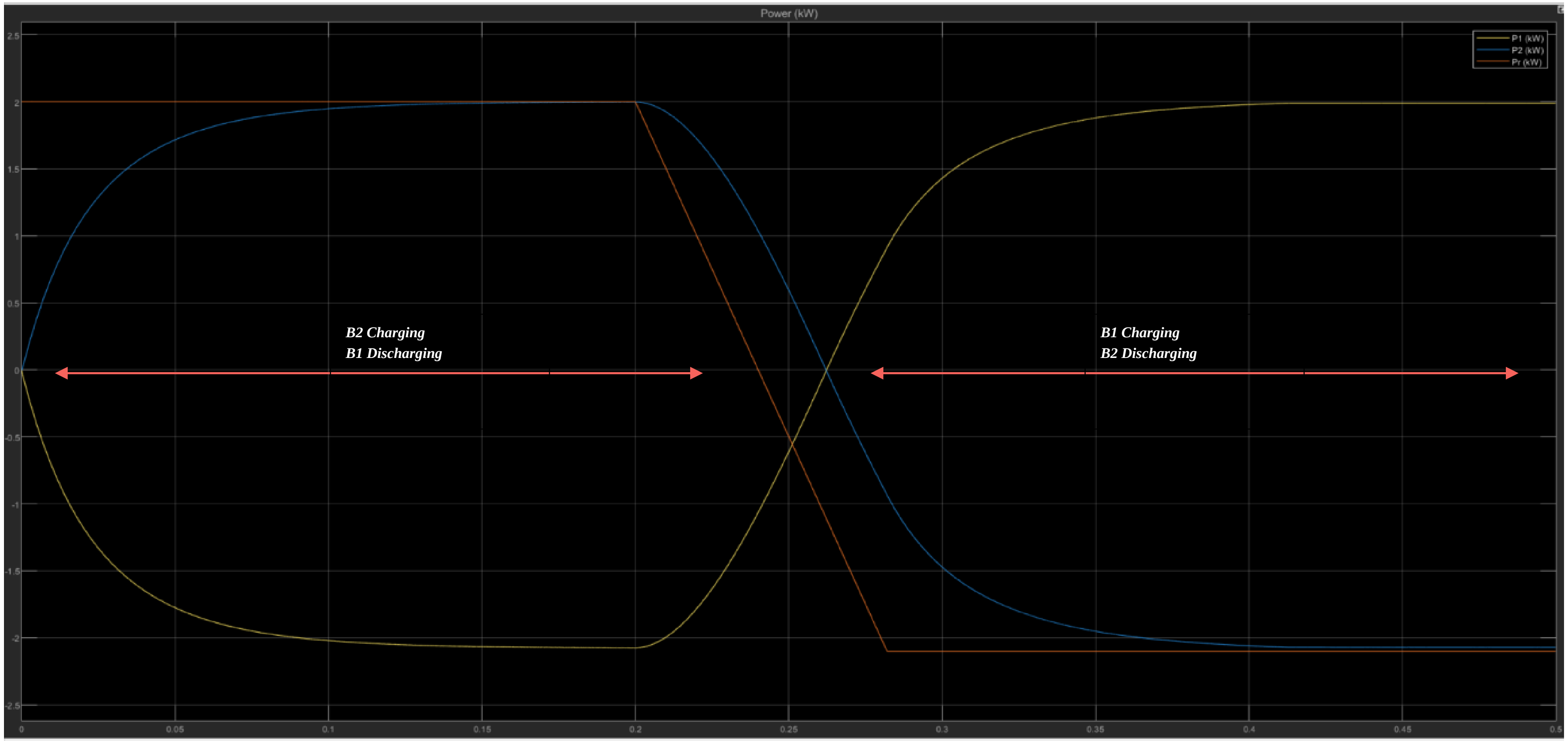
Equation 4 can be used to calculate the ripple Equation. 2.4% ripple current considered as a good percentage and met the specifications.

$$\%I_{ripple} = \frac{I_{max}-I_{min}}{I_{mean}} * 100\% \quad \rightarrow \frac{42.3-41.3}{41.7} * 100 = 2.4\% \quad (Equation\ 4)$$



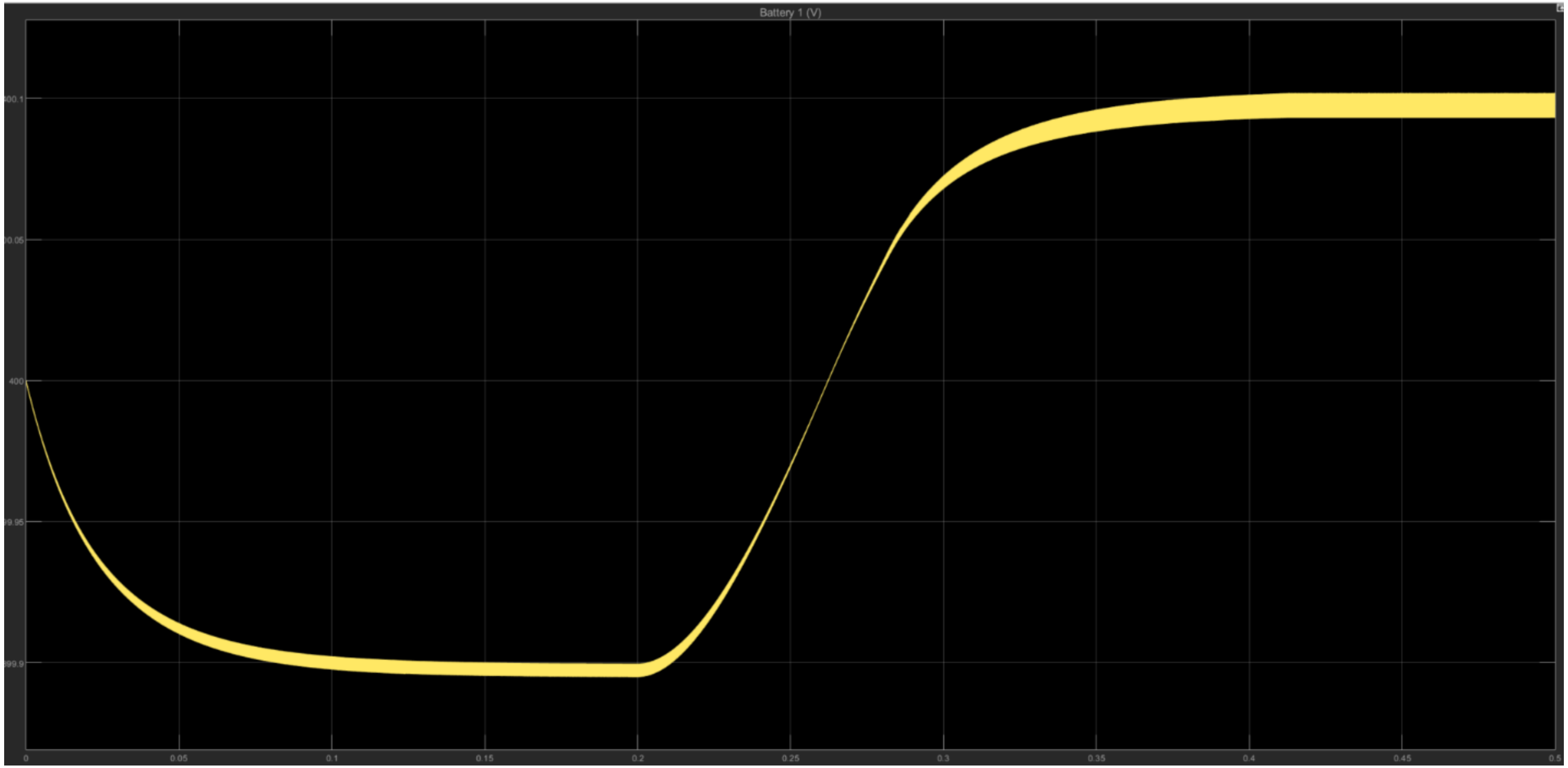
Simulation Results

Input/Output power curves



Simulation Results

Battery Voltage curve



Simulation Results

Equation 5 shows how to find the voltage ripple, the ripple is always less than 3% in both discharging and charging statues.

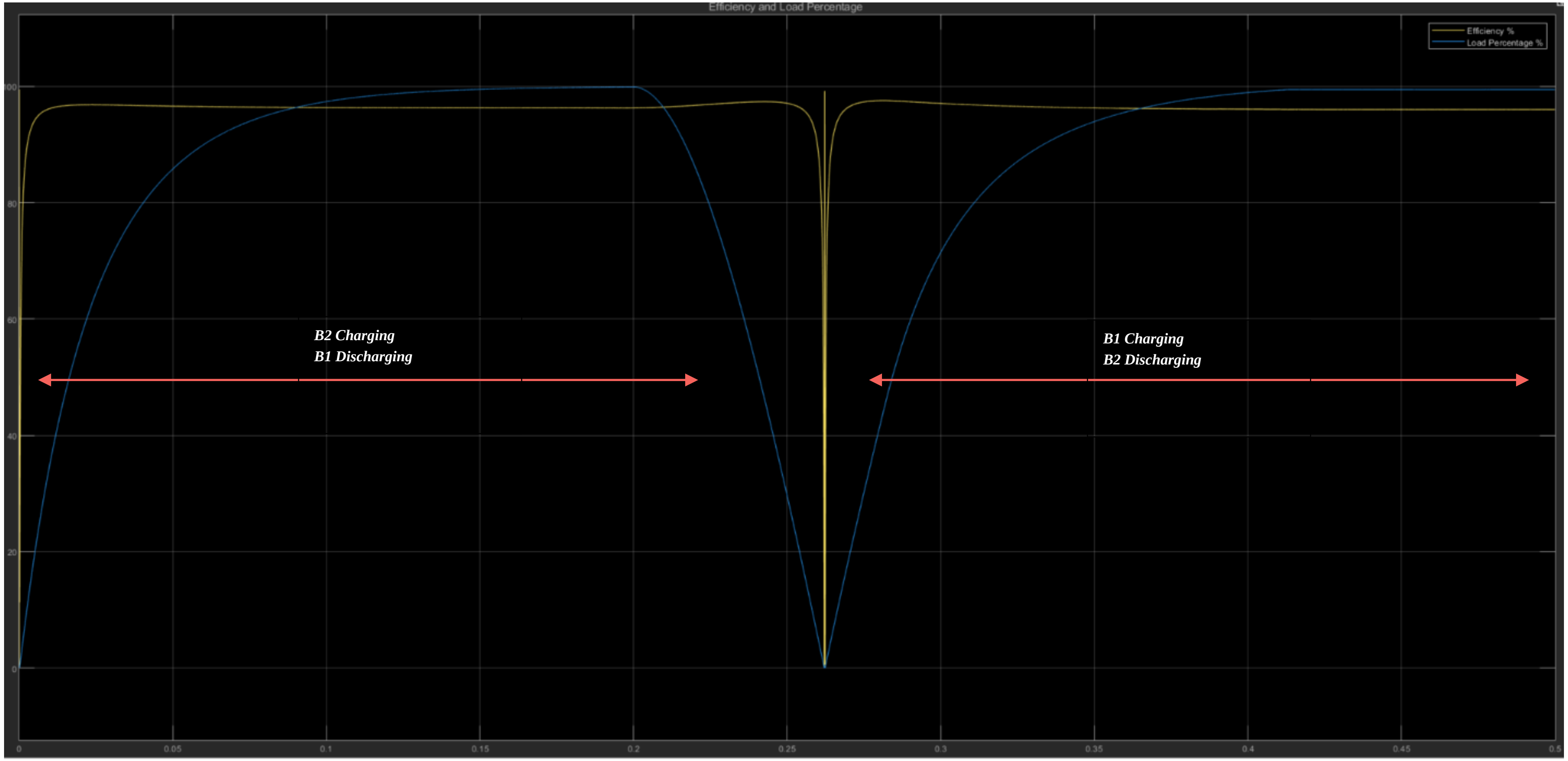
$$\%V_{ripple} = \frac{V_{max} - V_{min}}{V_{mean}} * 100\% \quad (Equation\ 5)$$

$$\%V_{ripple-discharging} = \frac{399.95 - 399.85}{399.889} * 100 = 0.025\%$$

$$\%V_{ripple-charging} = \frac{400.12 - 400.09}{400} * 100 = 0.007\%$$

Simulation Results

Power Efficiency curve



Simulation Results

At full load the efficiency equals 97%, using *Equation 6* the P_{loss} equals 0.0618 kW. These losses can be categorized into conduction loss, switching loss and transformer loss. $\%Efficiency = \frac{P_{out}}{P_{loss}+P_{out}} * 100\% \quad (Equation\ 6)$.

Simulation Results

Table 2 illustrates the loss types and their calculations.

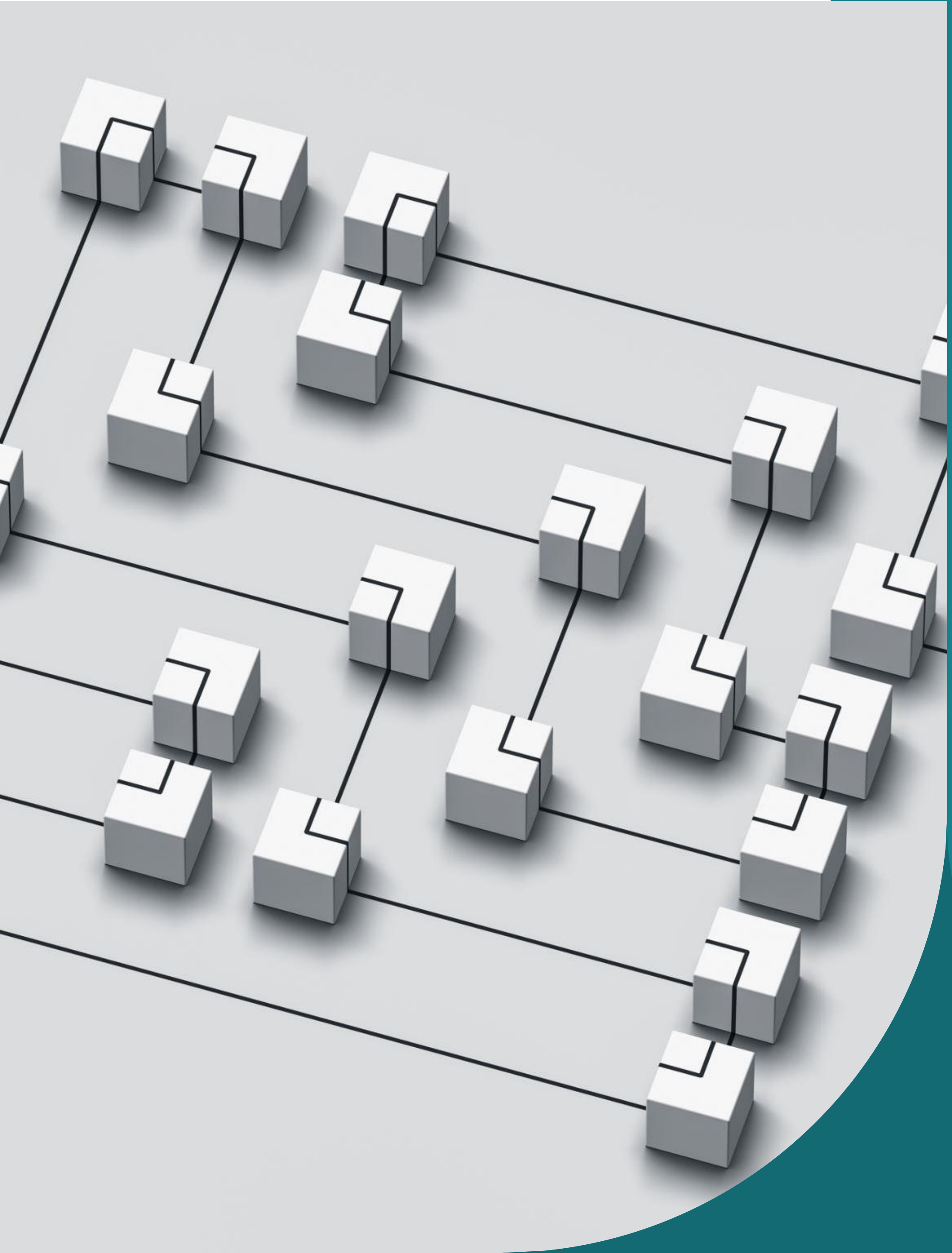
Power Loss	Equation	Value in W @ Full load
Conduction Sources Loss	$I^2 * R_{source}$	11.1
Conduction loss in transformers (primary side)	$I^2 * R_{primary}$	0.25
Conduction loss in transformers (secondary side)	$I^2 * R_{secondary}$	0.0018
Transformer core loss	$K * B^{\beta} * f$ | $I_m^2 * R_m$	3.16
MOSFET Conduction Loss	I^2 * Ron for each on mosfet	19
Diode Conduction Loss	$I^2 * Rd + I * V_d$ for each	22
Mosfet Switching Loss	$0.5 * V * I * f * T_{switching}$	2.5
Conduction loss in coupling inductor	$I^2 * R_L$	0.112
Conduction loss in DC link capacitor	$I_{ripple}^2 * R_c$	0.0026
Total Loss	$\sum P_{loss}$	60

Simulation Results

The simulation showed a good result that met the project requirements, *Table 3*

shows the project requirements and compares them with the results.

Project Requirements	Simulation Results
Current Ripple <4% @ Full load (Charging)	2.4%
Voltage Ripple <3% @ Full load	0.025%
Efficiency <95% @ Full load (Charging)	97%
Efficiency <92% @ 50 load (Charging)	96.7%
Efficiency <92% @ 25 load (Charging)	95%
Efficiency <95% @ Full load (discharging)	98%
Efficiency <92% @ 50 load (discharging)	97.5%
Efficiency <92% @ 25 load (discharging)	97%



Conclusion

- The simulation model of the proposed high-efficiency dual active bridge (DAB) converter proves its suitability and efficiency for distributed energy storage applications. The model gives an efficiency greater than 95% under all load conditions, with minimal voltage and current ripples, meeting the design requirements.

References

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Thank You! Any question?